KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · SCIENCE

Earth's Water by the Numbers

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 26):** Explain the difference between mass and weight. Then tell what a 10-kilogram bag weighs on Earth (9.8 newtons on each kilogram).

2. Predict: Out of every 100 cups of Earth's water, how many cups do you think are fresh water? Explain your guess.

3. Where do you think most of Earth's fresh water is: lakes, rivers, ice or underground? Why?

4. Name three ways your family uses water in one day. Which one do you think uses the most?

5. Draw a quick graph of anything you like to count (pets, books or rainy days). What kind of graph did you choose?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Counting Every Drop

Use evidence from today's lesson. Write in complete sentences.

1. **Compare.** Fresh groundwater is about 2,526,000 cubic miles; fresh lakes are about 21,830. Which is bigger, and about how many times bigger?

2. **Honest bar graph.** Graph 2015 U.S. water use: thermoelectric 133, irrigation 118, public supply 39 (billion gallons per day). Start at zero, label it and add a title.

3. **Two-step.** Turning off the tap while brushing saves about 8 gallons a day. How much for a family of 4 in one week?

4. **Find the mistake.** "Rivers hold most of Earth's fresh water." Explain what is wrong using the USGS numbers.

★ **Challenge: Explain.** How can the GRACE satellites tell that groundwater is shrinking, using gravity?

Looking Ahead: Next week: Heat and Thermal Energy. You will investigate how heat moves by conduction, convection and radiation, and test insulators.